Abhilekh Koirala

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Education

The Chinese University Of Hong Kong — Sha Tin, Hong Kong
Aerospace Science and Earth Information and Electrical and Computer Engineering — *Sept. 2024 – May 2028*

- **Relevant Coursework:** Problem Solving by Programming, Signals & Systems, Digital Logic and Systems, Computational Design & Automation, Robot Designing & Prototyping, Thermodynamics
- **Honors:** Honors at Entrance, Admissions Scholarship, Diversity & Inclusion Ambassador, Global Learning Citizen, Treasurer of South Asian Student Ambassador

Experience

Software Engineering Intern — June 2026 – August 2026
Oxbridge Economics — *Hong Kong*

- Built a PySpark/SQL pipeline in Databricks to analyze pest severity and efficacy across 3.6M+ turfgrass trial records, including ANOVA, Kruskal-Wallis, and Dunn's post-hoc testing
- Built the role-based access system (admin, site manager, rater) and invitation flow for a Strapi/PostgreSQL ops backend spanning 90+ API routes, deployed on AWS EC2/S3 with Docker
- Redesigned the Trial Designer into a canvas-based layout tool and rebuilt the admin analytics dashboard, both in React/Next.js
- Wrote the system's architecture docs (ERD, ADRs, a 181-row data dictionary across 24 entities) and built a interactive admin training manual

Research Assistant — March 2026 – June 2026
Physics Lab, CUHK — *Hong Kong*

- Built a Python pipeline that converts proprietary HORACE `.sqw` neutron-scattering files into NumPy arrays, with automatic fallback from Mantid's `LoadSQW/BinMD` to a custom binary parser when Mantid wasn't available
- Wrote a from-scratch binary reader for HORACE's raw pixel-table format (h, k, l, energy, signal, and error per neutron event) and rebinned it into a 4D Q-E histogram grid via `numpy.histogramdd`
- Processed inelastic neutron-scattering data on a Quantum Magnets sample across four incident energies (33, 49, 79, and 149 meV), enabling downstream ML-assisted spectral analysis
- Assisted in neutron spectroscopy experiments on an iron-oxide sample, gaining hands-on experience with scientific instrumentation

Analyst — June 2023 – January 2024
Himalayan Circuit — *Kathmandu, Nepal*

- Produced 3D route maps for 50+ hiking and trekking trails, visualizing distance, elevation gain/loss, and optimal routes
- Improved trekking-route recommendations by 15% by incorporating terrain difficulty and environmental factors
- Built a dataset covering 100+ km of trekking routes to support tourism-trend analytics and product planning
- Presented traveler carbon-emissions analysis findings at the United Nations Global Compact conference

Projects

NEPSEQuant | *React, Node.js, Python* — April 2026 – September 2026

- Applied persistent homology (Vietoris-Rips filtration, Betti numbers) on EWMA correlation matrices to classify NEPSE equity markets into four regimes (CALM, STRESS, CRISIS, FRAGMENTED)
- Built multi-horizon ensemble signals and risk-parity/minimum-risk-contribution position sizing on top of a walk-forward backtesting engine
- Modeled NEPSE-specific transaction costs: tiered broker commissions, exchange fees, and a 7.5% short-term capital gains tax across 71 rebalancing periods over a 6-year backtest
- Achieved a 1.47 Sharpe ratio, 2.52 Calmar ratio, and 385.71% net return

DocMD | *TypeScript, React, Vite, Chrome Extensions API* | GitHub — August 2026 – September 2026

- Built a Chrome extension that converts PDF, DOCX, PPTX, XLSX, CSV, HTML, and image files into structured Markdown for AI chatbots
- Added OCR support for scanned documents using Tesseract.js
- Handled multi-sheet spreadsheets, merged cells, and large-file splitting to stay under chatbot token limits

Skills

Programming Languages: Python, TypeScript, JavaScript, SQL, C, Verilog
Frameworks & Libraries: React, Node.js, Express, Flask, Vite, NumPy, pandas
Cloud & DevOps: AWS (EC2, S3, SES, IAM, CloudWatch), Docker, Databricks, PySpark, Git, GitHub
Scientific & Quantitative: Mantid, ANOVA, Kruskal-Wallis test, Dunn's post-hoc test, topological data analysis
GIS, CAD & Data Tools: SolidWorks, Advanced Excel, Google Sheets, Fatmap, Strava
Language: English, Nepali, Hindi